PriceWise AI:

Smart Shopping Intelligence & Real-Time Multi-Retailer Price Tracking Platform

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Problem Statement and Motivation

Fake discounts mislead online shoppers

Rapidly changing prices on major platforms cause confusion

Difficulty comparing prices across Amazon India, Flipkart, and Croma

Leads to poor decisions and extra spending



Proposed Solution, Main Objectives & Feature Scope

Real-time price comparison

Compare prices across multiple retailers instantly

Deal Score (0–100)

Evaluate deal quality with a score that ranges from 0 to 100

Price history visualization

Track price trends over time with clear visual insights

Price alerts

Get notified about significant price drops and deal opportunities



System Architecture and Workflow

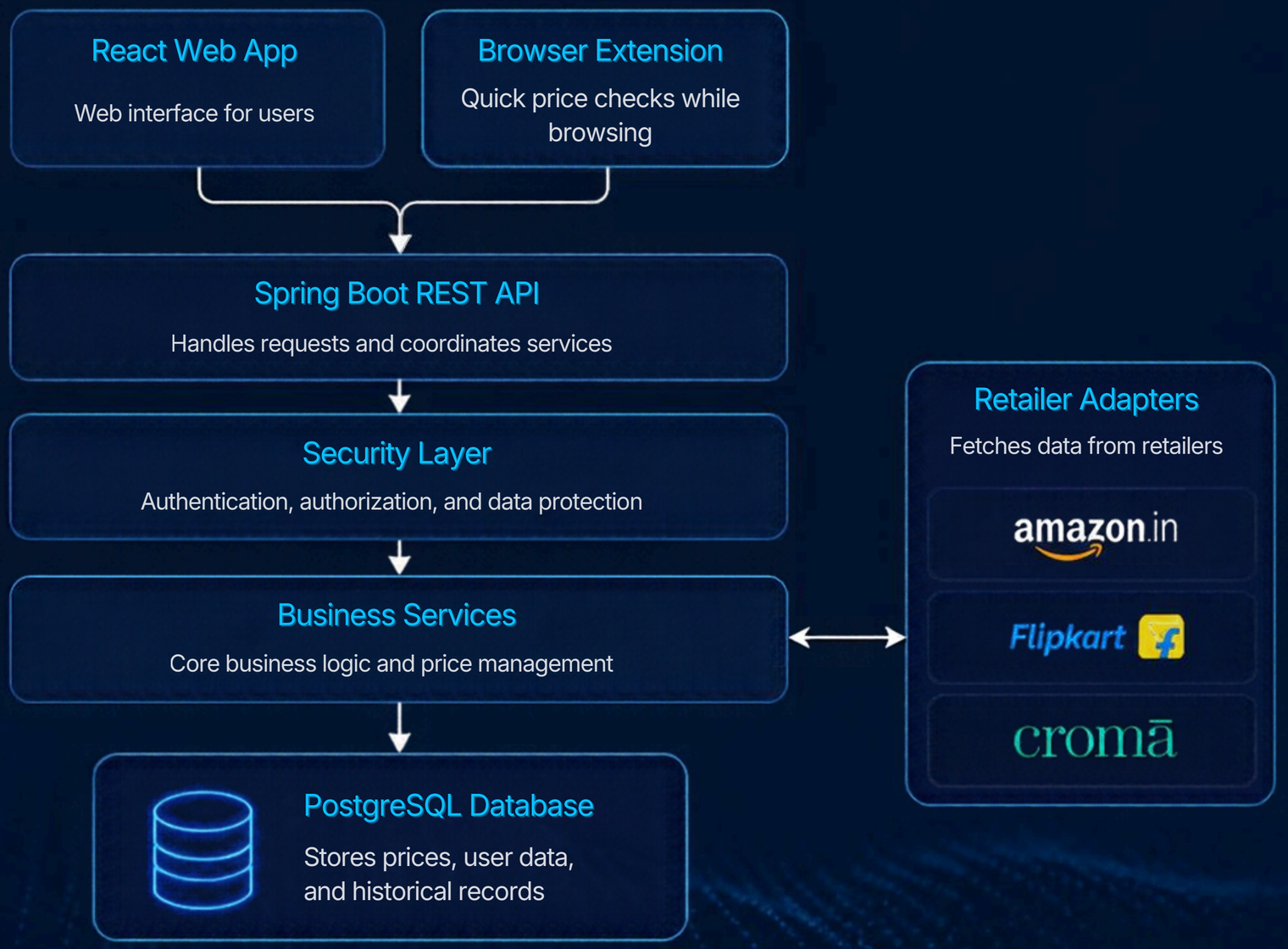
User interfaces connect shoppers to the platform

Backend services process requests and apply business logic

Security layer ensures safe and reliable operations

Retailer adapters fetch prices from multiple platforms

PostgreSQL stores price data and user information



Algorithm and Technology Stack

Historical Median Price
typical price based on past data

+

Discount Indicator
measures how much the price has dropped

+

Freshness Factor
assesses how recent the price update is

+

Deal Score (0–100)
combines all factors into a single score

→

Deal Score
0–100

Deal Score = $(w_1 \times \text{Historical Median}) + (w_2 \times \text{Discount Indicator}) + (w_3 \times \text{Freshness Factor})$

Technology Stack

modern, reliable, and scalable architecture

Frontend

React

TypeScript

Backend

Java

Spring Boot

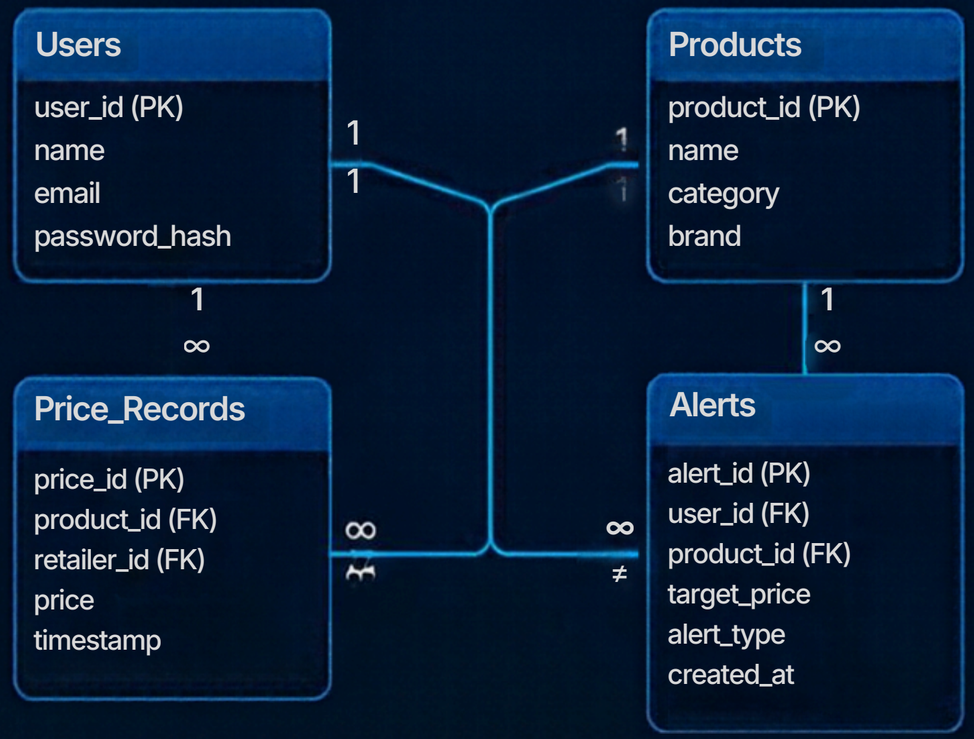
Database

PostgreSQL

Database Design, Security, Expected Benefits & Future Scope

Database Overview

Core Entities and Relationships



Security Considerations

Strong user authentication and access control
Data privacy and secure storage practices
Secure API access and encrypted communication

Expected Benefits

Better shopping decisions with price insights
Saves time by simplifying price comparison
Greater transparency in pricing across platforms

Future Scope

Expand coverage to more retailers and categories
Mobile app support for on-the-go access
AI-based personalized deal recommendations

Thank You

Questions & Answers

We appreciate your time and attention. We welcome your questions, feedback, and suggestions. Let’s discuss how we can make shopping smarter together.